

## Data Warehouse MCQ Quiz (20 Questions)

1. **Which of the following best defines a Data Warehouse?**  
A) A place to store transactional data  
**B) A system optimized for data analysis and reporting** ✓  
C) A real-time database for processing transactions  
D) A backup system for data recovery
2. **What is the primary difference between OLTP and OLAP?**  
A) OLTP is used for analytical processing, OLAP is for transactions  
**B) OLTP handles a large volume of transactions, OLAP handles complex queries** ✓  
C) OLTP supports decision-making, OLAP supports daily business operations  
D) OLAP updates data in real-time, OLTP does not
3. **In OLAP, what does "drill-down" analysis do?**  
A) Aggregates data into a summary view  
B) Filters out unnecessary data  
**C) Navigates from high-level data to more detailed data** ✓  
D) Joins data from multiple sources
4. **Which schema is considered the simplest in dimensional modeling?**  
A) Snowflake Schema  
**B) Star Schema** ✓  
C) Galaxy Schema  
D) Hybrid Schema
5. **A Data Lake primarily stores which type of data?**  
A) Structured data only  
**B) Unstructured and semi-structured data** ✓  
C) Only transactional data  
D) Data that is already cleaned and processed
6. **What is the main function of a Data Mart?**  
A) Stores transactional data  
**B) Is a smaller, department-specific version of a Data Warehouse** ✓  
C) Replaces a Data Warehouse  
D) Cleans and transforms data for reporting
7. **Which of the following statements is true about Star Schema?**  
A) It has multiple fact tables and no dimension tables  
B) It normalizes all dimension tables  
**C) It has a central fact table connected to denormalized dimension tables** ✓  
D) It is used only in OLTP systems
8. **The Snowflake Schema is different from the Star Schema because:**  
A) It does not have dimension tables  
**B) It normalizes dimension tables** ✓  
C) It only contains one fact table  
D) It is used only for real-time analytics

9. Which of these is NOT an OLAP operation?
- A) Roll-up
  - B) Drill-through
  - C) Slice and dice
  - D) Load balancing ✓
10. What is the primary benefit of a Slowly Changing Dimension (SCD)?
- A) It prevents changes in historical data
  - B) It allows tracking changes in dimension data over time ✓
  - C) It increases transaction speed in OLTP systems
  - D) It eliminates data redundancy
11. What is an ETL process in a Data Warehouse?
- A) Extract, Transform, Load ✓
  - B) Execute, Transfer, Log
  - C) Encrypt, Transform, Load
  - D) Extract, Transfer, Log
12. In a Data Warehouse, what is granularity?
- A) The level of detail stored in the data warehouse ✓
  - B) The size of the data warehouse storage
  - C) The number of indexes on a table
  - D) The frequency of ETL execution
13. Which Data Warehouse architecture allows multiple independent data marts to be integrated?
- A) Independent Data Mart Architecture ✗
  - B) Hub-and-Spoke Architecture ✓
  - C) Federated Architecture
  - D) Centralized Architecture
14. What type of OLAP system pre-aggregates data for fast query responses?
- A) MOLAP ✓
  - B) ROLAP
  - C) HOLAP
  - D) DOLAP
15. What is a key characteristic of a Fact Table?
- A) Contains only textual data
  - B) Stores aggregated numeric data ✓
  - C) Does not have foreign key relationships
  - D) Is always normalized
16. What is a conformed dimension in a Data Warehouse?
- A) A dimension used by multiple fact tables across different Data Marts ✓
  - B) A table with all numerical values
  - C) A dimension used only in OLTP
  - D) A duplicate of a fact table
17. In Data Warehouse, what is a surrogate key?
- A) A business key

- B) A primary key used in the source system
  - C) A system-generated unique identifier** ✓
  - D) A natural key from transactional data
18. **What does a Data Warehouse Bus Architecture use for consistency across multiple Data Marts?**
- A) Star Schema ✗
  - B) Conformed Dimensions** ✓
  - C) OLTP Database
  - D) NoSQL Databases
19. **Which technique helps improve query performance in Data Warehouses?**
- A) Using OLTP indexes ✗
  - B) Partitioning large tables** ✓
  - C) Removing fact tables
  - D) Normalizing all tables
20. **What is the role of metadata in a Data Warehouse?**
- A) It defines data structures, transformations, and lineage** ✓
  - B) It stores the actual data records ✗
  - C) It replaces transactional data
  - D) It is used only for security purposes

#### Informatica MCQ Quiz (20 Questions)

1. **What is the primary function of Informatica PowerCenter?**
- A) Real-time data processing
  - B) Data Integration and ETL operations** ✓
  - C) Transactional database management
  - D) Data visualization
2. **In Informatica, which transformation is used to perform row-wise calculations?**
- A) Expression Transformation** ✓
  - B) Aggregator Transformation
  - C) Joiner Transformation
  - D) Lookup Transformation
3. **What is the primary purpose of a Router Transformation?**
- A) To join data from multiple sources
  - B) To filter data based on multiple conditions** ✓
  - C) To perform row-wise calculations
  - D) To perform data aggregation
4. **Which of the following transformations is passive in Informatica?**
- A) Filter Transformation
  - B) Expression Transformation** ✓
  - C) Aggregator Transformation
  - D) Rank Transformation

5. **In Filter Transformation, what happens if the filter condition is FALSE?**  
A) The row is passed to the next transformation  
**B) The row is dropped from the pipeline** ✓  
C) The row is sent to the reject file  
D) The row is redirected to a secondary output
6. **What is the key difference between Joiner Transformation and SQL Joins?**  
A) Joiner Transformation allows joining only relational tables  
**B) Joiner Transformation can join heterogeneous sources** ✓  
C) SQL Joins allow joining data from multiple databases  
D) SQL Joins are always faster than Joiner Transformation
7. **Rank Transformation is used for:**  
A) Filtering records  
B) Sorting data  
**C) Assigning ranks to records based on a given condition** ✓  
D) Performing arithmetic operations
8. **What is the purpose of a Sequence Generator Transformation?**  
**A) Generates sequential numbers for primary keys** ✓  
B) Performs arithmetic calculations  
C) Sorts records based on ranking  
D) Filters duplicate records
9. **Which type of Lookup Transformation is faster in Informatica?**  
A) Connected Lookup  
B) Unconnected Lookup  
C) Dynamic Lookup  
**D) Cached Lookup** ✓
10. **In Slowly Changing Dimensions (SCD), which type maintains both historical and current data?**  
A) SCD Type 1  
**B) SCD Type 2** ✓  
C) SCD Type 3 ✗  
D) SCD Hybrid
11. **What is a session in Informatica?**  
A) A container for multiple workflows  
**B) A task that loads and transforms data in Informatica** ✓  
C) A transformation that joins data  
D) A query execution process
12. **What is the purpose of the Mapping Designer in Informatica?**  
A) To schedule workflows  
**B) To create and define ETL logic** ✓  
C) To manage database connections  
D) To monitor session logs

13. In Informatica, what is a repository? ☒
- A) A storage area for mappings, workflows, and sessions ☒
  - B) A backup system for session logs
  - C) A database storing source data
  - D) A container for SQL queries
14. What is the purpose of the Workflow Manager? ☒
- A) To create mappings
  - B) To execute ETL jobs and manage dependencies ☒
  - C) To filter data from source tables
  - D) To store historical data
15. How does an Aggregator Transformation improve performance? ☒
- A) By using sorted input ☒
  - B) By increasing the number of partitions
  - C) By performing row-wise calculations ☒
  - D) By reducing memory usage
16. Which transformation is used to replace NULL values with default values? ☒
- A) Filter Transformation
  - B) Aggregator Transformation
  - C) Expression Transformation ☒
  - D) Lookup Transformation
17. What does a Dynamic Lookup Cache do? ☒
- A) Stores lookup values in memory permanently
  - B) Updates lookup cache as new records are processed ☒
  - C) Clears lookup cache after each session
  - D) Performs lookup operations asynchronously
18. What happens when a session fails in Informatica? ☒
- A) The workflow is automatically restarted
  - B) The session stops and logs the error ☒
  - C) The session continues processing
  - D) The session ignores errors and loads data
19. Which of the following is true about Informatica PowerExchange? ☒
- A) It is used to integrate real-time data sources ☒
  - B) It is a reporting tool
  - C) It is used to design mappings
  - D) It is a data visualization tool
20. What is the purpose of Incremental Aggregation? ☒
- A) It performs aggregation on the entire dataset every time ☒
  - B) It processes only the new or changed data ☒
  - C) It duplicates records for better analysis
  - D) It ensures all data is aggregated in real-time

### Tableau MCQ Quiz (20 Questions with Correct Answers)

1. **What is the primary function of Tableau?**  
A) Data Integration  
B) Data Transformation  
**C) Data Visualization and Business Intelligence** ✓  
D) Database Management
2. **Which join type is NOT available in Tableau when joining tables?**  
A) Inner Join  
B) Full Outer Join  
**C) Cross Join** ✓  
D) Left Join
3. **Which chart type is best suited for showing trends over time?**  
A) Bar Chart  
**B) Line Chart** ✓  
C) Pie Chart  
D) Scatter Plot
4. **What is the difference between a Dimension and a Measure in Tableau?**  
A) Dimensions are numerical values, and Measures are categorical values  
**B) Dimensions contain qualitative data, while Measures contain quantitative data** ✓  
C) Measures store textual data, and Dimensions store numerical data  
D) There is no difference between Dimensions and Measures
5. **What does a Level of Detail (LOD) Expression do in Tableau?**  
**A) Summarizes data at different levels of granularity** ✓  
B) Filters data dynamically  
C) Changes the visualization type  
D) Modifies user permissions
6. **Which function is used for Table Calculations in Tableau?**  
A) SUM()  
**B) WINDOW\_SUM()** ✓  
C) COUNT()  
D) AVG()
7. **What is the purpose of a Dual Axis Chart in Tableau?**  
A) To combine two different charts in one visualization  
**B) To compare two measures with different scales** ✓  
C) To display dimensions only  
D) To merge two datasets
8. **What is the advantage of using Extracts instead of Live Connections in Tableau?**  
A) Extracts provide real-time data updates  
**B) Extracts improve performance by storing a snapshot of the data** ✓  
C) Extracts require an internet connection  
D) Extracts do not allow data manipulation

9. Which of the following functions is NOT a part of Tableau calculations?
- A) LOOKUP()
  - B) SUM()
  - C) TRANSFORM() ✓**
  - D) WINDOW\_AVG()
10. What is the purpose of Tableau Prep?
- A) To visualize data
  - B) To clean, shape, and prepare data before visualization ✓**
  - C) To store database records
  - D) To generate SQL queries
11. Which file format is used to save Tableau dashboards?
- A) .twb
  - B) .tds
  - C) .twbx ✓**
  - D) .tde ✗
12. In Tableau, what does the Pages shelf do?
- A) It filters data dynamically
  - B) It animates data changes over time ✓**
  - C) It connects multiple data sources ✗
  - D) It displays SQL queries
13. What type of visualization is best for comparing proportions?
- A) Line Chart
  - B) Heat Map
  - C) Pie Chart ✓**
  - D) Histogram
14. What is the role of a Calculated Field in Tableau?
- A) To filter data sources
  - B) To create custom calculations using formulas ✓**
  - C) To modify database tables
  - D) To store SQL queries
15. What is the default aggregation function for numeric fields in Tableau?
- A) SUM ✓**
  - B) AVG
  - C) COUNT
  - D) MAX
16. How does Tableau handle NULL values in data?
- A) It removes NULL values automatically
  - B) It treats NULL values as zero
  - C) It allows users to replace NULL values with default values ✓**
  - D) It ignores NULL values in all calculations

17. **What is the purpose of a Dashboard in Tableau?**
- A) To manage data sources
  - B) To create and publish reports
  - C) To combine multiple visualizations in a single view** ✓
  - D) To perform database queries
18. **In Tableau, what does the term "Blending" mean?**
- A) Combining multiple visualizations
  - B) Merging tables within the same data source
  - C) Joining multiple dashboards together
  - D) Combining data from different sources without joining** ✓
19. **Which of the following is an advantage of using Tableau Extensions?**
- A) They allow third-party tools to integrate with Tableau** ✓
  - B) They restrict visualization capabilities
  - C) They slow down the performance of dashboards
  - D) They only work with SQL databases
20. **What is the purpose of Story Points in Tableau?**
- A) To add narration to dashboards
  - B) To highlight key insights in a sequential manner** ✓
  - C) To filter data dynamically
  - D) To create calculated fields